Program : **Diploma in Fire Technology and Safety**	
Course Code: **4461**	Course Title: **Strength of Materials (FS)**
Semester:**4**	Credits: **4**
Course Category: **Program Core**	
Periods per week **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To understand basic concepts of stress and strain in materials.
- To learn how to calculate shear force and bending moments in beams.
- To explore how beams bend and twist under load.
- To learn about torsion in shafts, spring deflection, and the stresses in thin cylinders.

Course Prerequisites:

Topic	Course code	CourseName	Semester
Mathematics I & II		Mathematics I & II	1 & 2
Applied Physics I & II		Applied Physics I & II	1 & 2
Engineering Mechanics		Engineering Mechanics	2

CourseOutcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand basic stress and strain and calculate changes in dimensions of materials.	13	Understanding
CO2	Calculate shear force and bending moments, and draw diagrams for simple cases.	15	Applying
CO3	Understand the bending of beams, and solve problems related to deflection and columns.	16	Understanding
CO4	Learn the behavior of shafts, springs, and thin cylinders under load, and solve related problems.	14	Understanding
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	3					
CO4	3	3					

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand basic stress and strain and calculate changes in dimensions of materials**.		
M1.01	Learn types of forces and basic concepts of stress and strain.	2	Understanding
M1.02	Understand stress-strain diagrams for common materials.	2	Understanding
M1.03	Calculate stress and strain for uniform and composite sections.	2	Understanding
M1.04	Understand thermal stresses in materials..	2	Understanding
M1.05	Solve problems on stress and strain.	5	Applying
Contents: Types of stress and strain: tensile, compressive, and shear. Stress-strain diagrams for materials like Mild Steel (MS) and Cast Iron (CI).Thermal stress in materials			
CO2	**Calculate shear force and bending moments, and draw diagrams for simple cases.**		
M2.01	Understand different types of beams and loads.	3	Understanding
M2.02	Learn shear force and bending moment definitions	2	Understanding
M2.03	Calculate shear force and bending moments for various beam cases.	10	Applying

	Series Test 1	1	
Contents: Types of beams: cantilever, simply supported, overhanging, etc. Types of loads: point load, uniformly distributed load (UDL). Drawing shear force and bending moment diagrams.			
CO3	**Understand the bending of beams, and solve problems related to deflection and columns.**		
M3.01	Learn the basic theory of bending.	2	Understanding
M3.02	Understand the bending equation and apply it to simple problems.	3	Understanding
M3.03	Calculate bending stress, modulus of section, and moment of resistance.	6	Applying
M3.04	Understand beam deflection and apply standard formulae.	2	Understanding
M3.05	Learn about columns and struts, and calculate buckling load.	3	Applying
Contents: Bending stress and moment of resistance. Deflection of beams under point load and UDL.Columns: Euler's formula for buckling			
CO4	**Learn the behavior of shafts, springs, and thin cylinders under load, and solve related problems.**		
M4.01	Learn about shafts, calculate the polar moment of inertia (M.I.).	5	Understanding
M4.02	Understand how springs work and calculate their deflection and stiffness.	5	Understanding
M4.03	Understand stresses in thin cylindrical shells and calculate their thickness.	4	Understanding
	SeriesTest–II	1	

Contents:

Torsion in shafts: calculation of polar M.I. and stress.
Closed coil helical springs: stress, deflection, and stiffness.
Stresses in thin cylindrical shells subjected to internal pressure.

Text / Reference:

T/R	BookTitle/Author
T1	Strength of Materials by Dr. R.K. Bansal, Lakshmi Publishers
T2	Strength of Materials by R.S. Khurmi, S.Chand& Company Ltd.
R1	Strength of Materials by S.S. Bhavikatti, Vikas Publishing House
R2	Strength of Materials by Ramamrutham, Dhanpat Rai & Sons
R3	Strength of Materials by T.D. Gunneswara Rao, Cambridge University Press